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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 16th March Morning Shift

JEE Main Question Paper with Answer Key

16 Mar 2021	90	19	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	21 questions
Physics	28 questions
Out of syllabus	19 questions
Answer key	at the end

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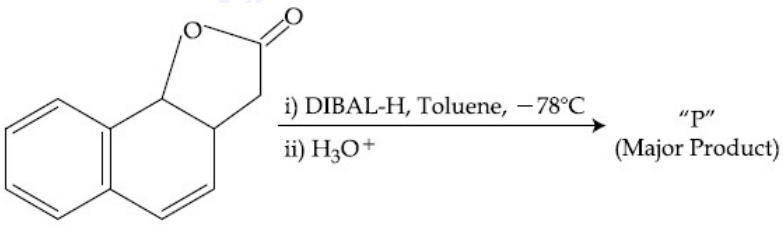
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Chemistry 22 questions

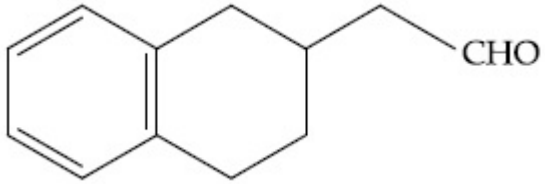
Q1. MCQ (Single Correct Answer)

Open solution

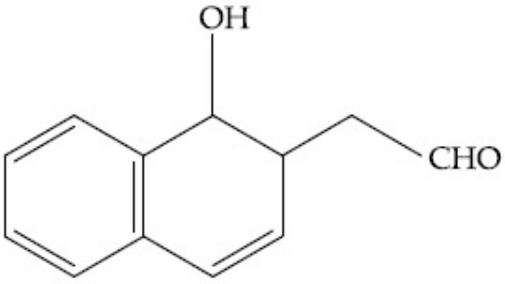


The product "P" in the above reaction is :

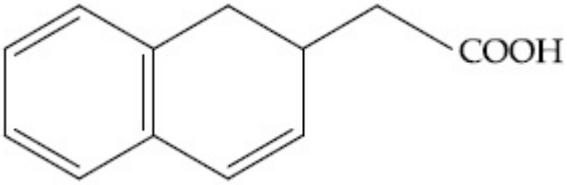
(A)



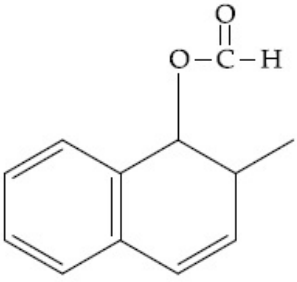
(B)



(C)



(D)



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Q2. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labeled as Assertion A and the other is labelled as Reason R :

Assertion A : The H—O—H bond angle in water molecule is 104.5°.

Reason R : The lone pair - lone pair repulsion of electrons is higher than the bond pair - bond pair repulsion.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both A and R are true, and R is the correct explanation of A

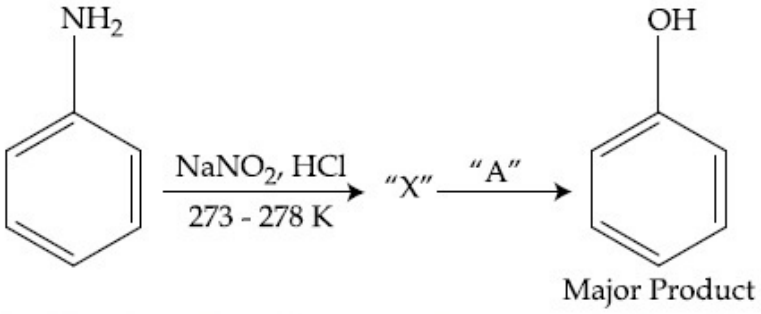
(B) A is true but R is false

(C) A is false but R is true

(D) Both A and R are true, but R is not the correct explanation of A

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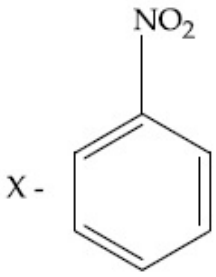
Q3. MCQ (Single Correct Answer)



Open solution

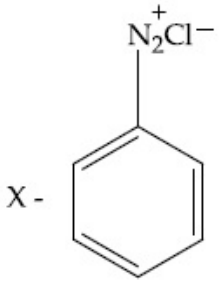
In the above chemical reaction, intermediate 'X' and reagent/condition 'A' are :

(A)



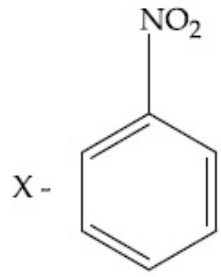
X - ; A - H_2O/Δ

(B)



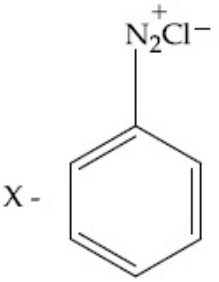
X - ; A - $H_2O/NaOH$

(C)



X - ; A - $H_2O/NaOH$

(D)



X - ; A - H_2O/Δ

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Q4. MCQ (Single Correct Answer)

Open solution

Assertion A : Enol form of acetone [CH_3COCH_3] exists in < 0.1% quantity. However, the enol form of acetyl acetone [$CH_3COCH_2OCCH_3$] exists in approximately 15% quantity.

Reason R : Enol form of acetyl acetone is stabilized by intramolecular hydrogen bonding, which is not possible in enol form of acetone.

Choose the correct statement :

(A) Both A and R are true but R is not the correct explanation of A

(B) A is true but R is false

(C) A is false but R is true

(D) Both A and R are true and R is the correct explanation of A

Q5. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : The E^o value for Ce^{4+} / Ce^{3+} is + 1.74 V.

Statement II : Ce is more stable in Ce^{4+} state than Ce^{3+} state.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is incorrect but statement II is correct

(B) Both statement I and statement II are correct

(C) Both statement I and statement II are incorrect

(D) Statement I is correct but statement II is incorrect

Q6. MCQ (Single Correct Answer)

Open solution

In chromotography technique, the purification of compound is independent of :

(A) Solubility of the compound

(B) Mobility or flow of solvent system

(C) Physical state of the pure compound

(D) Length of the column or TLC plate

Q7. MCQ (Single Correct Answer)

Open solution

An element of group 15, which is a metal and forms a hydride with strongest reducing power among group 15 hydrides. The element is :

(A) As

(B) Sb

(C) Bi

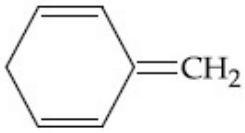
(D) P

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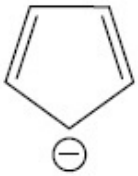
Q8. MCQ (Single Correct Answer)

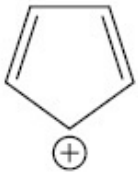
Open solution

Among the following, the aromatic compounds are :

(A) 

(B) 

(C) 

(D) 

Choose the correct answer from the following options :

(A) (B), (C) and (D) only

(B) (B) and (C) only

(C) (A), (B) and (C) only

(D) (A) and (B) only

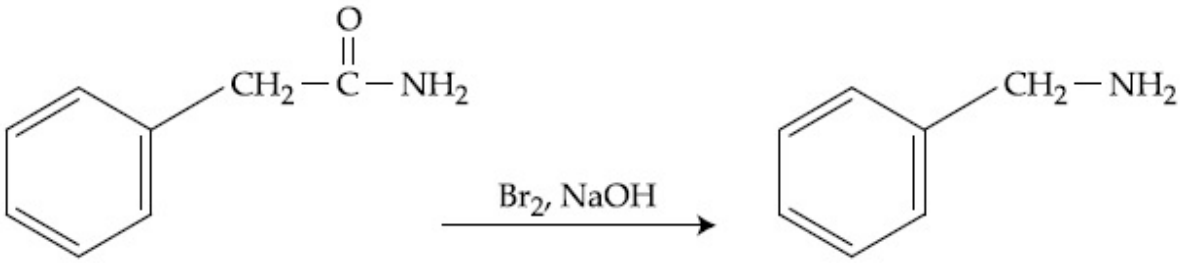
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Q9. MCQ (Single Correct Answer)

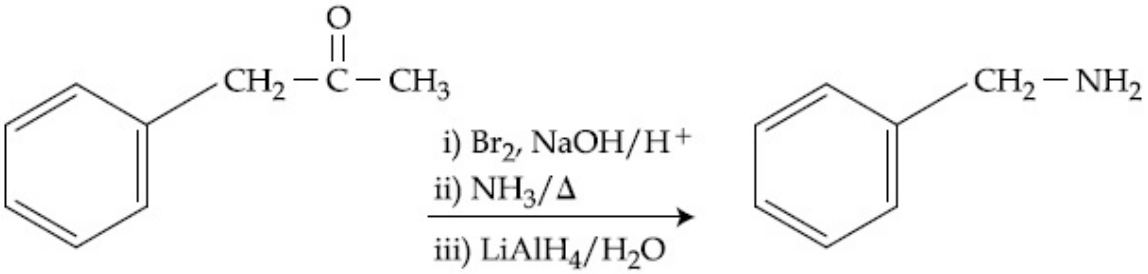
Open solution

Which of the following reactions DOES NOT involve Hofmann bromamide degradation?

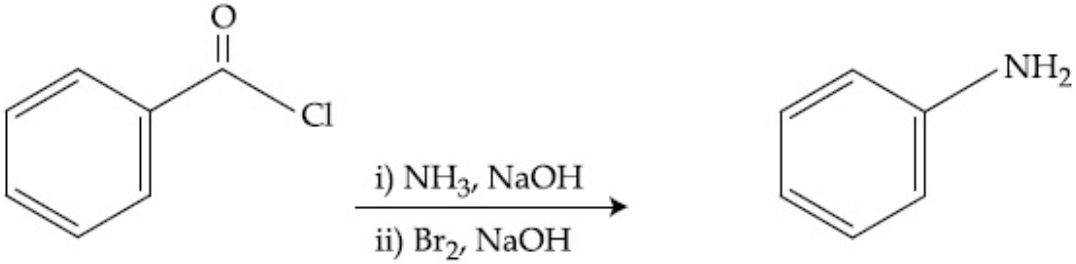
(A)



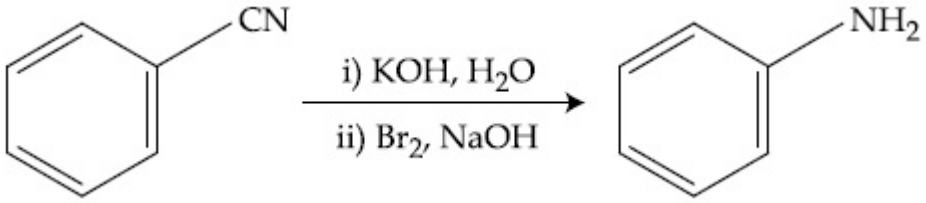
(B)



(C)



(D)



Q10. MCQ (Single Correct Answer)

Open solution

Which of the following is Lindlar catalyst?

(A) Partially deactivated palladised charcoal

(B) Zinc chloride and HCl

(C) Sodium and Liquid NH_3

(D) Cold dilute solution of $KMnO_4$

Q11. MCQ (Single Correct Answer)

Open solution

Which among the following pairs of Vitamins is stored in our body relatively for longer duration?

(A) Thiamine and Vitamin A

(B) Ascorbic acid and Vitamin D

(C) Thiamine and Ascorbic acid

(D) Vitamin A and Vitamin D

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Q12. MCQ (Single Correct Answer)

Open solution

Given below are two statement : one is labelled as Assertion A and the other is labelled as Reason R :

Assertion A : Size of Bk^{3+} ion is less than Np^{3+} ion.

Reason R : The above is a consequence of the lanthanoid contraction.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both A and B are true and R is the correct explanation of A

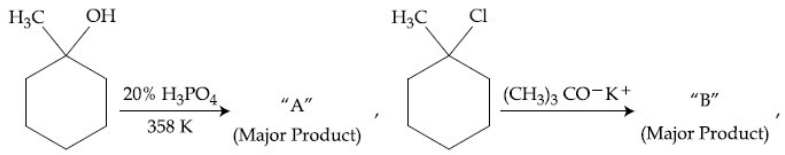
(B) A is false but R is true

(C) A is true but R is false

(D) Both A and B are true but R is not the correct explanation of A

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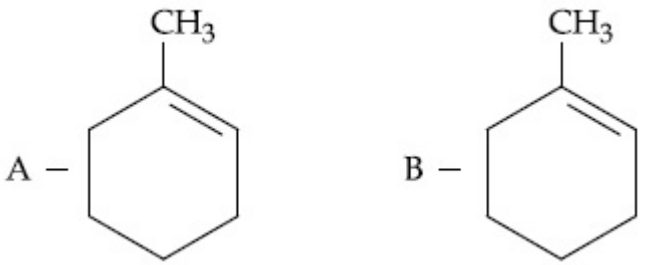
Q13. MCQ (Single Correct Answer)



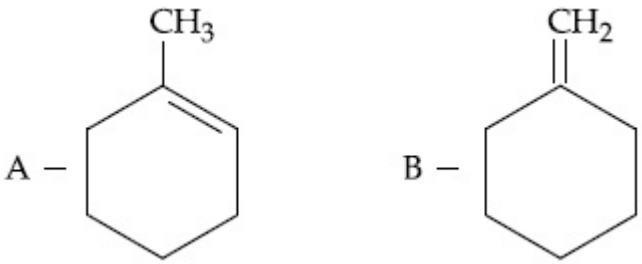
Open solution

The products 'A' and 'B' formed in above reactions are :

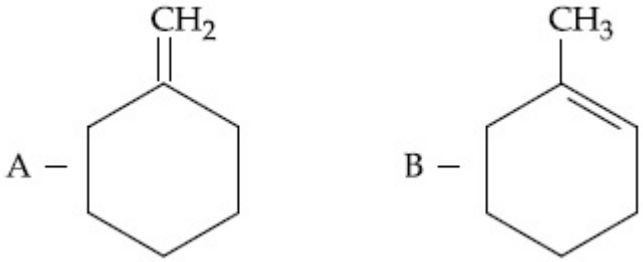
(A)



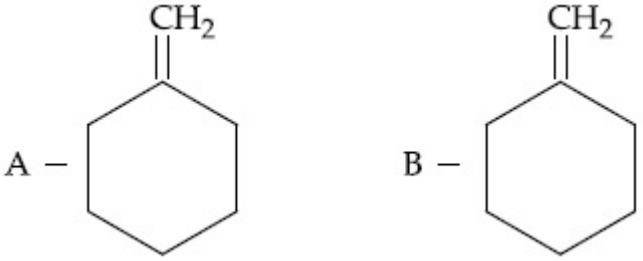
(B)



(C)



(D)



Q14. Numerical

The equivalents of ethylene diamine required to replace the neutral ligands from the coordination sphere of the trans-complex of $CoCl_3 \cdot 4NH_3$ is _____. (Round off to the Nearest Integer).

Open solution

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Q15. Numerical

Two salts A_2X and MX have the same value of solubility product of 4.0×10^{-12}. The ratio of their molar solubilities i.e. $\frac{S(A_2X)}{S(MX)}$ = _____. (Round off to the Nearest Integer)

Open solution

Q16. Numerical

Complete combustion of 750 g of an organic compound provides 420 g of CO_2 and 210 g of H_2O. The percentage composition of carbon and hydrogen in organic compound is 15.3 and _____ respectively. (Round off to the Nearest Integer).

Open solution

Q17. Numerical

For the reaction $A(g) \rightleftharpoons B(g)$ at 495 K, $\Delta_r G^\circ = -9.478$ kJ mol^{-1}.

If we start the reaction in a closed container at 495 K with 22 millimoles of A, the amount of B in the equilibrium mixture is _____ millimoles.

(Round off to the Nearest Integer). [R = 8.314 J mol^{-1} K^{-1}; ln 10 = 2.303]

Open solution

Q18. Numerical

$$2MnO_4^- + bC_2O_4^{2-} + cH^+ \rightarrow xMn^{2+} + yCO_2 + zH_2O$$

If the above equation is balanced with integer coefficients, the value of c is _____. (Round off to the Nearest Integer).

Open solution

Q19. Numerical

A 6.50 molal solution of KOH (aq.) has a density of 1.89 g cm^{-3}. The molarity of the solution is _____ mol dm^{-3}. (Round off to the Nearest Integer).

[Atomic masses : K : 39.0 u; O : 16.0 u; H : 1.0 u]

Open solution

Q20. Numerical

AB_2 is 10% dissociated in water to A^{2+} and B^-. The boiling point of a 10.0 molal aqueous solution of AB_2 is _____°C. (Round off to the Nearest Integer).

[Given : Molal elevation constant of water K_b = 0.5 K kg mol^{-1}, boiling point of pure water = 100°C]

Open solution

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Q21. Numerical

When light of wavelength 248 nm falls on a metal of threshold energy 3.0 eV, the de-Broglie wavelength of emitted electrons is _____ $\overset{o}{A}$. (Round off to the Nearest Integer).

[Use : $\sqrt{3}$ = 1.73, h = 6.63 $\times$ 10^{-34} Js

m_e = 9.1 $\times$ 10^{-31} kg; c = 3.0 $\times$ 10^{8} ms^{-1}; 1eV = 1.6 $\times$ 10^{-19} J]

Open solution

Q22. Numerical

The decomposition of formic acid on gold surface follows first order kinetics. If the rate constant at 300 K is 1.0 $\times$ 10^{-3} s^{-1} and the activation energy E_a = 11.488 kJ mol^{-1}, the rate constant at 200 K is _____ $\times$ 10^{-5} s^{-1}. (Round off to the Nearest Integer). (Given : R = 8.314 J mol^{-1} K^{-1})

Open solution

Mathematics 21 questions

Q23. MCQ (Single Correct Answer)

Let a vector $\alpha\hat{i} + \beta\hat{j}$ be obtained by rotating the vector $\sqrt{3}\hat{i} + \hat{j}$ by an angle 45° about the origin in counterclockwise direction in the first quadrant. Then the area of triangle having vertices (α, β), (0, β) and (0, 0) is equal to :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{2}$ (C) 1 (D) $2\sqrt{2}$

Open solution

Q24. MCQ (Single Correct Answer)

Let $S_k = \sum_{r=1}^{k} \tan^{-1}\left(\frac{6^r}{2^{2r+1}+3^{2r+1}}\right)$. Then $\lim_{k\to\infty} S_k$ is equal to :

(A) $\cot^{-1}\left(\frac{3}{2}\right)$ (B) $\frac{\pi}{2}$

(C) $\tan^{-1}(3)$ (D) $\tan^{-1}\left(\frac{3}{2}\right)$

Open solution

Q25. MCQ (Single Correct Answer)

Let the position vectors of two points P and Q be $3\hat{i} - \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$, respectively. Let R and S be two points such that the direction ratios of lines PR and QS are (4, −1, 2) and (−2, 1, −2), respectively. Let lines PR and QS intersect at T. If the vector $\overrightarrow{TA}$ is perpendicular to both $\overrightarrow{PR}$ and $\overrightarrow{QS}$ and the length of vector $\overrightarrow{TA}$ is $\sqrt{5}$ units, then the modulus of a position vector of A is :

(A) $\sqrt{171}$ (B) $\sqrt{227}$ (C) $\sqrt{482}$ (D) $\sqrt{5}$

Open solution

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Q26. MCQ (Single Correct Answer)

The number of elements in the set $\{x \in R : (|x| - 3)\,|x + 4| = 6\}$ is equal to :

(A) 4 (B) 2 (C) 3 (D) 1

Open solution

Q27. MCQ (Single Correct Answer)

Let a complex number z, $|z| \neq 1$,

satisfy $\log_{\frac{1}{\sqrt{2}}}\left(\frac{|z|+11}{(|z|-1)^2}\right) \leq 2$. Then, the largest value of |z| is equal to _____.

(A) 5 (B) 8 (C) 6 (D) 7

Open solution

Q28. MCQ (Single Correct Answer)

Let $A = \begin{bmatrix} i & -i \\ -i & i \end{bmatrix}, i = \sqrt{-1}$. Then, the system of linear equations $A^8 \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 64 \end{bmatrix}$ has :

(A) Exactly two solutions (B) Infinitely many solutions

(C) A unique solution (D) No solution

Open solution

Q29. MCQ (Single Correct Answer)

If n is the number of irrational terms in the expansion of $\left(3^{1/4} + 5^{1/8}\right)^{60}$, then (n − 1) is divisible by :

(A) 30 (B) 8 (C) 7 (D) 26

Open solution

Q30. MCQ (Single Correct Answer)

The locus of the midpoints of the chord of the circle, $x^2 + y^2 = 25$ which is tangent to the hyperbola, $\frac{x^2}{9} - \frac{y^2}{16} = 1$ is :

(A) $(x^2 + y^2)^2 - 9x^2 + 16y^2 = 0$ (B) $(x^2 + y^2)^2 - 9x^2 + 144y^2 = 0$

(C) $(x^2 + y^2)^2 - 16x^2 + 9y^2 = 0$ (D) $(x^2 + y^2)^2 - 9x^2 - 16y^2 = 0$

Open solution

Q31. MCQ (Single Correct Answer)

Let the functions f : R → R and g : R → R be defined as :

$$f(x) = \begin{cases} x + 2, & x < 0 \\ x^2, & x \geq 0 \end{cases} \text{ and}$$

$$g(x) = \begin{cases} x^3, & x < 1 \\ 3x - 2, & x \geq 1 \end{cases}$$

Then, the number of points in R where (fog) (x) is NOT differentiable is equal to :

(A) 0 (B) 3 (C) 1 (D) 2

Open solution

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Q32. MCQ (Single Correct Answer)

Consider three observations a, b, and c such that b = a + c. If the standard deviation of a + 2, b + 2, c + 2 is d, then which of the following is true?

(A) $b^2 = 3(a^2 + c^2) + 9d^2$

(B) $b^2 = 3(a^2 + c^2) - 9d^2$

(C) $b^2 = 3(a^2 + c^2 + d^2)$

(D) $b^2 = a^2 + c^2 + 3d^2$

Open solution

Q33. MCQ (Single Correct Answer)

The range of $a \in R$ for which the

function $f(x) = (4a - 3)(x + \log_e 5) + 2(a - 7)\cot\left(\frac{x}{2}\right)\sin^2\left(\frac{x}{2}\right)$, $x \neq 2n\pi$, $n \in N$ has critical points, is :

(A) $[1, \infty)$

(B) $(-3, 1)$

(C) $\left[-\frac{4}{3}, 2\right]$

(D) $(-\infty, -1]$

Open solution

Q34. MCQ (Single Correct Answer)

If for $x \in \left(0, \frac{\pi}{2}\right)$, $\log_{10}\sin x + \log_{10}\cos x = -1$ and $\log_{10}(\sin x + \cos x) = \frac{1}{2}(\log_{10} n - 1)$, $n > 0$, then the value of n is equal to :

(A) 16

(B) 9

(C) 12

(D) 20

Open solution

Q35. MCQ (Single Correct Answer)

A pack of cards has one card missing. Two cards are drawn randomly and are found to be spades. The probability that the missing card is not a spade, is :

(A) $\frac{39}{50}$

(B) $\frac{3}{4}$

(C) $\frac{22}{425}$

(D) $\frac{52}{867}$

Open solution

Q36. MCQ (Single Correct Answer)

If y = y(x) is the solution of the differential equation,

$\frac{dy}{dx} + 2y\tan x = \sin x, y\left(\frac{\pi}{3}\right) = 0$, then the maximum value of the function y(x) over R is equal to:

(A) $\frac{1}{8}$

(B) 8

(C) $-\frac{15}{4}$

(D) $\frac{1}{2}$

Open solution

Q37. Numerical

Let the curve y = y(x) be the solution of the differential equation, $\frac{dy}{dx} = 2(x + 1)$. If the numerical value of area bounded by the curve y = y(x) and x-axis is $\frac{4\sqrt{8}}{3}$, then the value of y(1) is equal to _____.

Open solution

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Q38. Numerical

Let f : R $\rightarrow$ R be a continuous function such that f(x) + f(x + 1) = 2, for all x$\in$R.

If $I_1 = \int_0^8 f(x)dx$ and $I_2 = \int_{-1}^3 f(x)dx$, then the value of $I_1 + 2I_2$ is equal to _____.

Open solution

Q39. Numerical

If the normal to the curve y(x) = $\int_0^x (2t^2 - 15t + 10)dt$ at a point (a, b) is parallel to the line x + 3y = -5, a > 1, then the value of | a + 6b | is equal to _____.

Open solution

Q40. Numerical

Consider an arithmetic series and a geometric series having four initial terms from the set {11, 8, 21, 16, 26, 32, 4}. If the last terms of these series are the maximum possible four digit numbers, then the number of common terms in these two series is equal to _____.

Open solution

Q41. Numerical

If $\lim_{x\to 0} \frac{ae^x - b\cos x + ce^{-x}}{x\sin x} = 2$, then a + b + c is equal to _____.

Open solution

Q42. Numerical

Let z and ω be two complex numbers such that $\omega = z\bar{z} - 2z + 2$, $\left|\frac{z+i}{z-3i}\right| = 1$ and Re(ω) has minimum value. Then, the minimum value of n $\in$ N for which ω^n is real, is equal to _____.

Open solution

Q43. Numerical

The total number of 3 $\times$ 3 matrices A having entries from the set {0, 1, 2, 3} such that the sum of all the diagonal entries of AA^T is 9, is equal to _____.

Open solution

Physics 28 questions

Q44. MCQ (Single Correct Answer)

The pressure acting on a submarine is 3 $\times$ 10^5 Pa at a certain depth. If the depth is doubled, the percentage increase in the pressure acting on the submarine would be :

(Assume that atmospheric pressure is 1 $\times$ 10^5 Pa and density of water is 10^3 kg m^{-3}, g = 10 ms^{-2})

(A) $\frac{200}{5}$ % (B) $\frac{200}{3}$ % (C) $\frac{3}{200}$ % (D) $\frac{5}{200}$ %

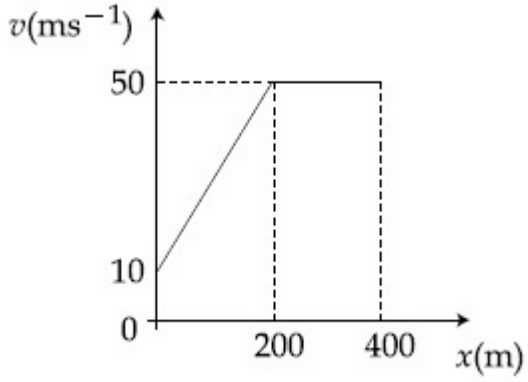
Open solution

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Q45. MCQ (Single Correct Answer)

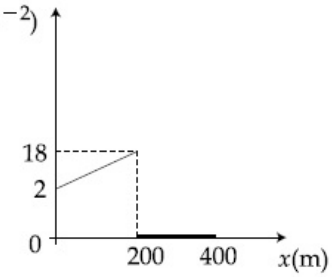
Open solution

The velocity-displacement graph describing the motion of bicycle is shown in the figure.

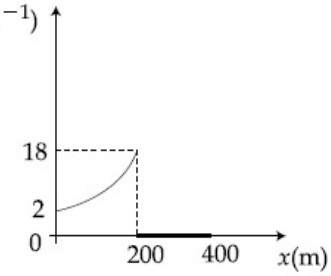


The acceleration-displacement graph of the bicycle's motion is best described by :

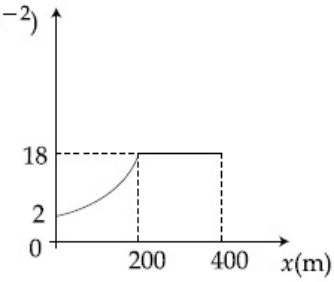
(A)



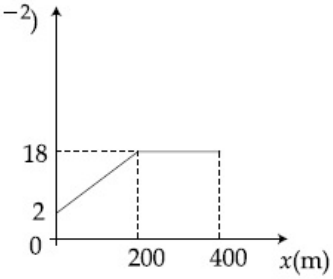
(B)



(C)



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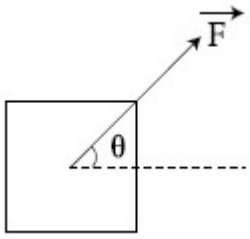
(D) 

Q46. MCQ (Single Correct Answer)

Open solution

A block of mass m slides along a floor while a force of magnitude F is applied to it at an angle θ as shown in figure. The coefficient of kinetic friction is μ_k. Then, the block's acceleration 'a' is given by :

(g is acceleration due to gravity)



(A) $\frac{F}{m}\cos\theta + \mu_K\left(g - \frac{F}{m}\sin\theta\right)$

(B) $\frac{F}{m}\cos\theta - \mu_K\left(g - \frac{F}{m}\sin\theta\right)$

(C) $-\frac{F}{m}\cos\theta - \mu_K\left(g - \frac{F}{m}\sin\theta\right)$

(D) $\frac{F}{m}\cos\theta - \mu_K\left(g + \frac{F}{m}\sin\theta\right)$

Q47. MCQ (Single Correct Answer)

Open solution

A block of 200 g mass moves with a uniform speed in a horizontal circular groove, with vertical side walls of radius 20 cm. If the block takes 40 s to complete one round, the normal force by the side walls of the groove is :

(A) 9.859×10^{-2} N (B) 0.0314 N (C) 9.859×10^{-4} N (D) 6.28×10^{-3} N

Q48. MCQ (Single Correct Answer)

Open solution

Time period of a simple pendulum is T inside a lift when the lift is stationary. If the lift moves upwards with an acceleration g/2, the time period of pendulum will be :

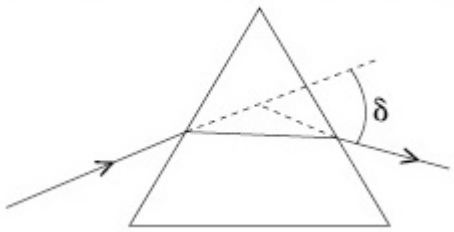
(A) $\sqrt{3}T$ (B) $\sqrt{\frac{2}{3}}T$ (C) $\frac{T}{\sqrt{3}}$ (D) $\sqrt{\frac{3}{2}}T$

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Q49. MCQ (Single Correct Answer)

Open solution

The angle of deviation through a prism is minimum when



(A) Incident ray and emergent ray are symmetric to the prism.

(B) The refracted ray inside the prism becomes parallel to its base

(C) Angle of incidence is equal to that of the angle of emergence

(D) When angle of emergence is double the angle of incidence

Choose the correct answer from the options given below :

(A) Statements (B) and (C) are true

(B) Only statements (A) and (B) are true

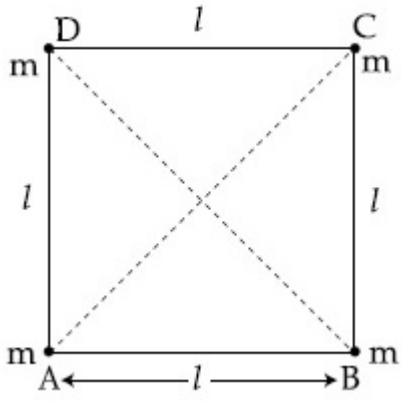
(C) Statements (A), (B) and (C) are true

(D) Only statement (D) is true

Q50. MCQ (Single Correct Answer)

Open solution

Four equal masses, m each are placed at the corners of a square of length (l) as shown in the figure. The moment of inertia of the system about an axis passing through A and parallel to DB would be :



(A) $\sqrt{3}\,ml^2$

(B) $2\,ml^2$

(C) ml^2

(D) $3\,ml^2$

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Q51. MCQ (Single Correct Answer)

In thermodynamics, heat and work are :

(A) Path functions

(B) Point functions

(C) Extensive thermodynamics state variables

(D) Intensive thermodynamic state variables

Open solution

Q52. MCQ (Single Correct Answer)

For changing the capacitance of a given parallel plate capacitor, a dielectric material of dielectric constant K is used, which has the same area as the plates of the capacitor. The thickness of the dielectric slab is $\frac{3}{4}$d, where 'd' is the separation between the plates of parallel plate capacitor. The new capacitance (C') in terms of original capacitance (C_0) is given by the following relation :

(A) $C' = \frac{3+K}{4K}C_0$

(B) $C' = \frac{4+K}{3}C_0$

(C) $C' = \frac{4K}{K+3}C_0$

(D) $C' = \frac{4}{3+K}C_0$

Open solution

Q53. MCQ (Single Correct Answer)

A plane electromagnetic wave of frequency 500 MHz is travelling in vacuum along y-direction. At a particular point in space and time,

$\overrightarrow{B} = 8.0 \times 10^{-8}\,\hat{z}$T. The value of electric field at this point is :

(speed of light = 3×10^8 ms^{-1})

$\hat{x}$, $\hat{y}$, $\hat{z}$ are unit vectors along x, y and z directions.

(A) 2.6 $\hat{x}$ V/m

(B) $-24\,\hat{x}$ V/m

(C) $24\,\hat{x}$ V/m

(D) $-2.6\,\hat{y}$ V/m

Open solution

Q54. MCQ (Single Correct Answer)

The maximum and minimum distances of a comet from the Sun are 1.6×10^{12} m and 8.0×10^{10} m respectively. If the speed of the comet at the nearest point is 6×10^4 ms^{-1}, the speed at the farthest point is :

(A) 3.0×10^3 m/s

(B) 6.0×10^3 m/s

(C) 1.5×10^3 m/s

(D) 4.5×10^3 m/s

Open solution

Q55. MCQ (Single Correct Answer)

A conducting wire of length 'l', area of cross-section A and electric resistivity ρ is connected between the terminals of a battery. A potential difference V is developed between its ends, causing an electric current.

If the length of the wire of the same material is doubled and the area of cross-section is halved, the resultant current would be :

(A) $4\frac{VA}{\rho l}$

(B) $\frac{3}{4}\frac{VA}{\rho l}$

(C) $\frac{1}{4}\frac{VA}{\rho l}$

(D) $\frac{1}{4}\frac{\rho l}{VA}$

Open solution

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Q56. MCQ (Single Correct Answer)

For an electromagnetic wave travelling in free space, the relation between average energy densities due to electric (U_e) and magnetic (U_m) fields is :

(A) $U_e = U_m$ (B) $U_e \neq U_m$ (C) $U_e < U_m$ (D) $U_e > U_m$

Open solution

Q57. MCQ (Single Correct Answer)

The stopping potential in the context of photoelectric effect depends on the following property of incident electromagnetic radiation :

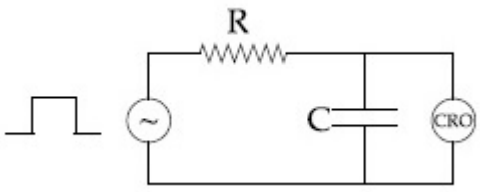
(A) Phase (B) Frequency (C) Amplitude (D) Intensity

Open solution

Q58. MCQ (Single Correct Answer)

An RC circuit as shown in the figure is driven by a AC source generating a square wave. The output wave pattern monitored by CRO would look close to :

Open solution



(A) 

(B) 

(C) 

(D) 

Q59. MCQ (Single Correct Answer)

One main scale division of a vernier callipers is 'a' cm and n^{th} division of the vernier scale coincide with $(n-1)^{th}$ division of the main scale. The least count of the callipers in mm is :

(A) $\frac{10a}{n}$ (B) $\frac{10na}{(n-1)}$

(C) $\left(\frac{n-1}{10n}\right)a$ (D) $\frac{10a}{(n-1)}$

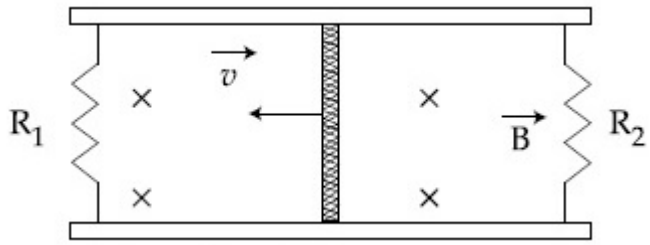
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Q60. MCQ (Single Correct Answer)

Open solution

A conducting bar of length L is free to slide on two parallel conducting rails as shown in the figure



Two resistors R_1 and R_2 are connected across the ends of the rails. There is a uniform magnetic field $\vec{B}$ pointing into the page. An external agent pulls the bar to the left at a constant speed v.

The correct statement about the directions of induced currents I_1 and I_2 flowing through R_1 and R_2 respectively is :

(A) Both I_1 and I_2 are in clockwise direction

(B) I_1 is in clockwise direction and I_2 is in anticlockwise direction

(C) I_1 is in anticlockwise direction and I_2 is in clockwise direction

(D) Both I_1 and I_2 are in anticlockwise direction

Q61. MCQ (Single Correct Answer)

Open solution

The volume V of an enclosure contains a mixture of three gases, 16 g of oxygen, 28 g of nitrogen and 44 g of carbon dioxide at absolute temperature T. Consider R as universal gas constant. The pressure of the mixture of gases is :

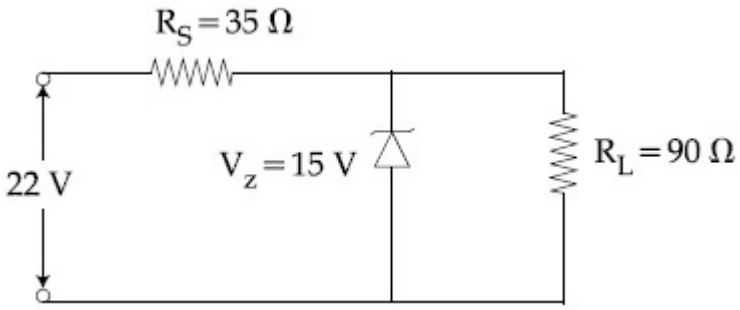
(A) $\frac{3RT}{V}$ (B) $\frac{4RT}{V}$ (C) $\frac{88RT}{V}$ (D) $\frac{5}{2}\frac{RT}{V}$

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Q62. Numerical

The value of power dissipated across the zener diode (V_z = 15V) connected in the circuit as shown in the figure is x $\times$ 10^{-1} watt.

Open solution



The value of x, to the nearest integer, is _____.

Q63. Numerical

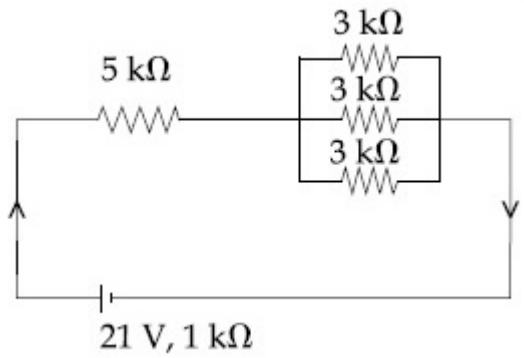
A fringe width of 6 mm was produced for two slits separated by 1 mm apart. The screen is placed 10 m away. The wavelength of light used is 'x' nm. The value of 'x' to the nearest integer is _____.

Open solution

Q64. Numerical

In the figure given, the electric current flowing through the 5 kΩ resistor is 'x' mA.

Open solution



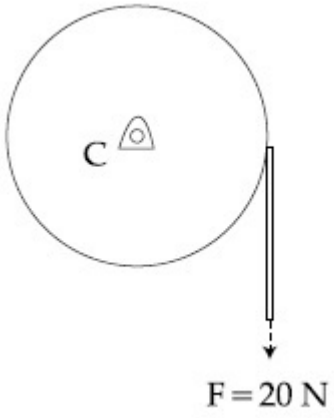
The value of x to the nearest integer is _____.

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Q65. Numerical

Consider a 20 kg uniform circular disk of radius 0.2 m. It is pin supported at its center and is at rest initially. The disk is acted upon by a constant force F = 20 N through a massless string wrapped around its periphery as shown in the figure.

Open solution



Suppose the disk makes n number of revolutions to attain an angular speed of 50 rad s^{-1}.

The value of n, to the nearest integer, is _____.

[Given : In one complete revolution, the disk rotates by 6.28 rad]

Q66. Numerical

The resistance R = $\frac{V}{I}$, where V = (50 $\pm$ 2)V and I = (20 $\pm$ 0.2)A. The percentage error in R is 'x'%. The value of 'x' to the nearest integer is _____.

Open solution

Q67. Numerical

A sinusoidal voltage of peak value 250 V is applied to a series LCR circuit, in which R = 8Ω, L = 24 mH and C = 60 μF. The value of power dissipated at resonant condition is 'x' kW. The value of x to the nearest integer is _____.

Open solution

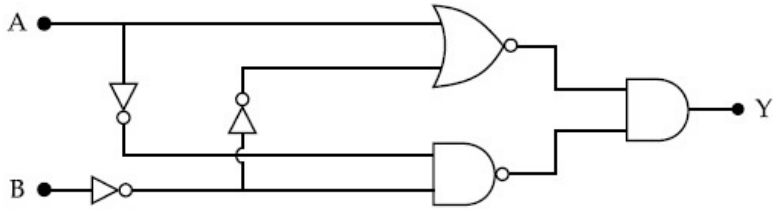
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Q68. Numerical

In the logic circuit shown in the figure, if input A and B are 0 and 1 respectively, the output at Y would be 'x'.

Open solution

The value of x is _____.



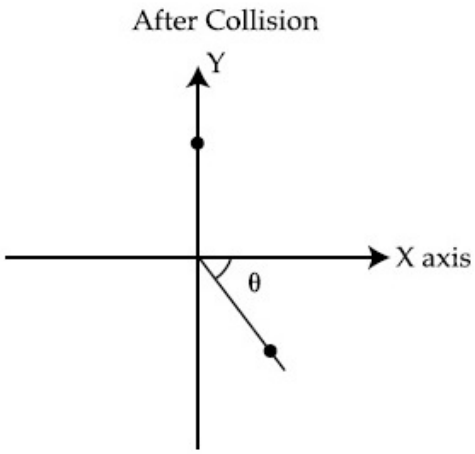
Q69. Numerical

A ball of mass 10 kg moving with a velocity $10\sqrt{3}$ m s^{-1} along X-axis, hits another ball of mass 20 kg which is at rest. After collision, the first ball comes to rest and the second one disintegrates into two equal pieces. One of the pieces starts moving along Y-axis at a speed of 10 m/s. The second piece starts moving at a speed of 20 m/s at an angle θ (degree) with respect to the X-axis.

Open solution

The configuration of pieces after collision is shown in the figure.

The value of θ to the nearest integer is _____.



Q70. Numerical

The first three spectral lines of H-atom in the Balmer series are given λ_1, λ_2, λ_3 considering the Bohr atomic model, the wave lengths of first and third spectral lines $\left(\frac{\lambda_1}{\lambda_3}\right)$ are related by a factor of approximately 'x' $\times 10^{-1}$.

Open solution

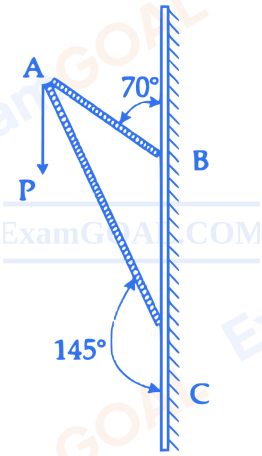
The value of x, to the nearest integer, is _____.

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Q71. Numerical

Consider a frame that is made up of two thin massless rods AB and AC as shown in the figure. A vertical force $\vec{P}$ of magnitude 100 N is applied at point A of the frame.

Open solution



Suppose the force $\vec{P}$ is resolved parallel to the arms AB and AC of the frame.

The magnitude of the resolved component along the arm AC is xN.

The value of x, to the nearest integer, is _____.

[Given : sin(35°) = 0.573, cos(35°) = 0.819

sin(110°) = 0.939, cos(110°) = – 0.342]

Out of syllabus 19 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q72. MCQ (Single Correct Answer) · **Chemistry**

The type of pollution that gets increased during the day time and in the presence of O_3 is :

(A) Oxidising smog (B) Global warming (C) Acid rain (D) Reducing smog

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

The functions of antihistamine are :

(A) Antiallergic and Analgesic (B) Antacid and antialergic

(C) Analgesic and antacid (D) Antiallergic and antidepressant

Open solution

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Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Match List - I with List - II :

	List - I Industrial process		**List - II Application**
(a)	Haber's process	(i)	HNO_3 synthesis
(b)	Ostwald's process	(ii)	Aluminium extraction
(c)	Contact process	(iii)	NH_3 synthesis
(d)	Hal-Heroult process	(iv)	H_2SO_4 synthesis

Choose the correct answer from the options given below :

(A) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(B) (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)

(C) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Q75. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Given below are two statements :

Statement I : H_2O_2 can act as both oxidising and reducing agent in basic medium.

Statement II : In the hydrogen economy, the energy is transmitted in the form of dihydrogen.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both statement I and statement II are false

(B) Statement I is false but statement II is true

(C) Statement I is true but statement II is false

(D) Both statement I and statement II are true

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Q76. MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II :

	List - I **Name of oxo acid**		**List - II** **Oxidation state of 'P'**
(a)	Hypophosphorous acid	(i)	+5
(b)	Orthophosphoric acid	(ii)	+4
(c)	Hypophosphoric acid	(iii)	+3
(d)	Orthophosphorous acid	(iv)	+2
		(v)	+1

Open solution

Choose the correct answer from the options given below :

(A) (a) - (iv), (b) - (v), (c) - (ii), (d) - (iii)

(B) (a) - (iv), (b) - (i), (c) - (ii), (d) - (iii)

(C) (a) - (v), (b) - (i), (c) - (ii), (d) - (iii)

(D) (a) - (v), (b) - (iv), (c) - (ii), (d) - (iii)

Q77. MCQ (Single Correct Answer) · **Chemistry**

Given below are two statements :

Statement I : Both $CaCl_2 . 6H_2O$ and $MgCl_2 . 8H_2O$ undergo dehydration on heating.

Statement II : BeO is amphoteric whereas the oxides of other elements in the same group are acidic.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both statement I and statement II are false

(B) Statement I is false but statement II is true

(C) Statement I is true but statement II is false

(D) Both statement I and statement II are true

Open solution

Q78. MCQ (Single Correct Answer) · **Chemistry**

The process that involves the removal of sulphur from the ores is :

(A) Roasting **(B)** Smelting **(C)** Leaching **(D)** Refining

Open solution

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Q79. Numerical · **Chemistry**

A certain element crystallises in a bcc lattice of unit cell edge length $27\overset{o}{A}$. If the same element under the same conditions crystallises in the fcc lattice, the edge length of the unit cell in $\overset{o}{A}$ will be _____. (Round off to the Nearest Integer).

[Assume each lattice point has a single atom]

[Assume $\sqrt{3}$ = 1.73, $\sqrt{2}$ = 1.41]

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

Which of the following Boolean expression is a tautology?

(A) $(p \wedge q) \vee (p \rightarrow q)$ (B) $(p \wedge q) \vee (p \vee q)$

(C) $(p \wedge q) \rightarrow (p \rightarrow q)$ (D) $(p \wedge q) \wedge (p \rightarrow q)$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

If the three normals drawn to the parabola, $y^2 = 2x$ pass through the point (a, 0) a $\neq 0$, then 'a' must be greater than :

(A) $\frac{1}{2}$ (B) 1 (C) -1 (D) $-\frac{1}{2}$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

Let P be a plane lx + my + nz = 0 containing

the line, $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$. If plane P divides the line segment AB joining

points A$(-3, -6, 1)$ and B(2, 4, -3) in ratio k : 1 then the value of k is equal to :

(A) 2 (B) 3 (C) 1.5 (D) 4

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

The number of roots of the equation, $(81)^{\sin^2 x} + (81)^{\cos^2 x} = 30$ in the interval $[0, \pi]$ is equal to :

(A) 2 (B) 3 (C) 4 (D) 8

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

If for a > 0, the feet of perpendiculars from the points A$(a, -2a, 3)$ and B(0, 4, 5) on the plane lx + my + nz = 0 are points C$(0, -a, -1)$ and D respectively, then the length of line segment CD is equal to :

(A) $\sqrt{41}$ (B) $\sqrt{55}$ (C) $\sqrt{31}$ (D) $\sqrt{66}$

Open solution

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Q85. MCQ (Single Correct Answer) · **Mathematics**

Let [x] denote greatest integer less than or equal to x. If for n∈N,

$$(1-x+x^3)^n=\sum_{j=0}^{3n}a_jx^j,$$

then $\sum_{j=0}^{\left[\frac{3n}{2}\right]}a_{2j}+4\sum_{j=0}^{\left[\frac{3n-1}{2}\right]}a_{2j+1}$ is equal to :

(A) 2^n-1 (B) n (C) 2 (D) 1

Open solution

Q86. Numerical · **Mathematics**

Let ABCD be a square of side of unit length. Let a circle C_1 centered at A with unit radius is drawn. Another circle C_2 which touches C_1 and the lines AD and AB are tangent to it, is also drawn. Let a tangent line from the point C to the circle C_2 meet the side AB at E. If the length of EB is $\alpha+\sqrt{3}\,\beta$, where α, β are integers, then $\alpha+\beta$ is equal to _____.

Open solution

Q87. Numerical · **Mathematics**

Let f : (0, 2) → R be defined as f(x) = $\log_2\left(1+\tan\left(\frac{\pi x}{4}\right)\right)$. Then, $\lim_{n\to\infty}\frac{2}{n}\left(f\left(\frac{1}{n}\right)+f\left(\frac{2}{n}\right)+\ldots+f(1)\right)$ is equal to _____.

Open solution

Q88. Numerical · **Mathematics**

Let $P=\begin{bmatrix}-30 & 20 & 56\\ 90 & 140 & 112\\ 120 & 60 & 14\end{bmatrix}$ and

$A=\begin{bmatrix}2 & 7 & \omega^2\\ -1 & -\omega & 1\\ 0 & -\omega & -\omega+1\end{bmatrix}$ where

$\omega=\frac{-1+i\sqrt{3}}{2}$, and I_3 be the identity matrix of order 3. If the determinant of the matrix $(P^{-1}AP-I_3)^2$ is $\alpha\omega^2$, then the value of α is equal to _____.

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

A 25 m long antenna is mounted on an antenna tower. The height of the antenna tower is 75 m. The wavelength (in meter) of the signal transmitted by this antenna would be :

(A) 200 (B) 400 (C) 100 (D) 300

Open solution

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Q90. MCQ (Single Correct Answer) · **Physics**

Open solution

A bar magnet of length 14 cm is placed in the magnetic meridian with its north pole pointing towards the geographic north pole. A neutral point is obtained at a distance of 18 cm from the center of the magnet. If B_H = 0.4 G, the magnetic moment of the magnet is : (1G = 10^{-4} T)

(A) 2.880×10^2 J T^{-1}

(B) 2.880 J T^{-1}

(C) 2.880×10^3 J T^{-1}

(D) 28.80 J T^{-1}

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Answer Key

Chemistry

1. **B**	2. **A**	3. **D**	4. **D**	5. **D**
6. **C**	7. **C**	8. **B**	9. **B**	10. **A**
11. **D**	12. **C**	13. **B**	14. **2**	15. **50**
16. **3**	17. **20**	18. **16**	19. **9**	20. **106**
21. **9**	22. **10**			

Mathematics

23. **B**	24. **A**	25. **A**	26. **B**	27. **D**
28. **D**	29. **D**	30. **A**	31. **C**	32. **B**
33. **C**	34. **C**	35. **A**	36. **A**	37. **2**
38. **16**	39. **406**	40. **3**	41. **4**	42. **4**
43. **766**				

Physics

44. **B**	45. **A**	46. **B**	47. **C**	48. **B**
49. **C**	50. **D**	51. **A**	52. **C**	53. **B**
54. **A**	55. **C**	56. **A**	57. **B**	58. **D**
59. **A**	60. **B**	61. **D**	62. **5**	63. **600**
64. **3**	65. **20**	66. **5**	67. **4**	68. **0**
69. **30**	70. **15**	71. **82**		

Out of syllabus

72. **A**	73. **B**	74. **B**	75. **D**	76. **C**
77. **A**	78. **A**	79. **33**	80. **C**	81. **B**
82. **A**	83. **C**	84. **D**	85. **D**	86. **1**
87. **1**	88. **36**	89. **C**	90. **B**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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